

## ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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# 1. GATE CH 2025

## A. Formulae

### 1.1.1 Dot Product of Two Vectors

- Matrix and Summation Forms: For two  $n$ -dimensional vectors  $\mathbf{a}$  and  $\mathbf{b}$ ,

$$\mathbf{a}^\top \mathbf{b} = \sum_{i=1}^n a_i b_i \quad (1.1.1.1)$$

### 1.1.2 Argument of Product of Complex Numbers

- Euler Form: For  $z_1 = r_1 e^{i\theta_1}$  and  $z_2 = r_2 e^{i\theta_2}$ ,

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2} \quad (1.1.2.1)$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)} \quad (1.1.2.2)$$

$$\arg(z_1 z_2) = \theta_1 + \theta_2 \pmod{2\pi} \quad (1.1.2.3)$$

$$\theta_1 + \theta_2 = \arg(z_1 z_2) + 2n\pi, \quad n \in \mathbb{Z} \quad (1.1.2.4)$$

### 1.1.3 Mean Residence Time from RTD

- Mean Residence Time: For a residence time distribution  $E(t)$ ,

$$\bar{t} = \frac{\int_0^\infty t E(t) dt}{\int_0^\infty E(t) dt} \quad (1.1.3.1)$$

$$\int_0^\infty E(t) dt = 1 \quad (1.1.3.2)$$

$$\bar{t} = \int_0^\infty t E(t) dt \quad (1.1.3.3)$$

### 1.1.4 Permutations and Combinations

- Number of Ways of Choosing  $r$  Items from  $n$  Items:

$${}^n C_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} \quad (1.1.4.1)$$

### 1.1.5 Expectation of a Function

- Continuous Random Variable: For a continuous random variable  $X$  with probability density function  $f_X(x)$ ,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad (1.1.5.1)$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (1.1.5.2)$$

- Discrete Random Variable:

$$\mathbb{E}[g(X)] = \sum_x g(x) \Pr(X = x) \quad (1.1.5.3)$$

### 1.1.6 Newton-Raphson Method

- First-Order Recurrence: For a differentiable function  $f(x)$ , the root is iteratively approximated using the Newton-Raphson recurrence

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.1.6.1)$$

### 1.1.7 Eigenvalues, Eigenvectors and Trace

- Definition: For a square matrix  $\mathbf{A}$ ,

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}, \quad \mathbf{x} \neq \mathbf{0} \quad (1.1.7.1)$$

where  $\lambda$  is an eigenvalue and  $\mathbf{x}$  is the corresponding eigenvector.

- Characteristic Equation:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \quad (1.1.7.2)$$

- Relation with Trace: If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the eigenvalues of  $\mathbf{A} \in \mathbb{R}^{n \times n}$ ,

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i = \lambda_1 + \lambda_2 + \dots + \lambda_n \quad (1.1.7.3)$$

Also, from the eigenvalue product relation,

$$\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i \quad (1.1.7.4)$$

### 1.1.8 Trapezoidal Method for Plotting Differential Equations

- Differential Equation:

$$\frac{dy}{dx} = f(x, y) \quad (1.1.8.1)$$

- Trapezoidal Update:

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})] \quad (1.1.8.2)$$

- The method approximates the solution by using the average of the slopes at two consecutive points. The calculated points can then be used to plot the numerical solution curve.

### 1.1.9 Jacobian Matrix and Divergence

- Jacobian Matrix: For the vector field  $\mathbf{F} = (F_1 \ F_2 \ \dots \ F_n)^\top$  with variables  $\mathbf{x} = (x_1 \ x_2 \ \dots \ x_n)^\top$ ,

$$J_{\mathbf{F}} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \dots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \dots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1} & \frac{\partial F_n}{\partial x_2} & \dots & \frac{\partial F_n}{\partial x_n} \end{pmatrix} \quad (1.1.9.1)$$

- Divergence:

$$\nabla \cdot \mathbf{F} = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i} \quad (1.1.9.2)$$

- Relation Between Jacobian and Divergence:

$$\nabla \cdot \mathbf{F} = \text{tr}(J_{\mathbf{F}}) \quad (1.1.9.3)$$

#### 1.1.10 Dead Time and Time Delay in the Laplace Domain

- Time Delay: A pure time delay of  $\theta$  in the time domain is represented in the Laplace domain by

$$e^{-\theta s} \quad (1.1.10.1)$$

- Delayed Signal: If a signal  $x(t)$  is delayed by  $\theta$ , then

$$x_d(t) = x(t - \theta)u(t - \theta) \quad (1.1.10.2)$$

$$X_d(s) = e^{-\theta s}X(s) \quad (1.1.10.3)$$

- Significance in a Transfer Function: For a process transfer function

$$G(s) = G_0(s)e^{-\theta s}, \quad (1.1.10.4)$$

the factor  $e^{-\theta s}$  represents a pure time delay or dead time of  $\theta$  time units, while  $G_0(s)$  represents the process dynamics without the delay.

- The dead time shifts the process response along the time axis by  $\theta$  without changing the steady-state gain. For a first-order plus dead-time model,

$$G_{\text{FOPDT}}(s) = \frac{Ke^{-\theta s}}{\tau s + 1} \quad (1.1.10.5)$$

where  $K$  is the process gain,  $\tau$  is the time constant, and  $\theta$  is the dead time. This is the same FOPDT form given in (1.1.11.1).

#### 1.1.11 Maximum Slope Method for FOPDT Approximation

- First-Order Plus Dead-Time Model:

$$G(s) = \frac{Ke^{-\theta s}}{\tau s + 1} \quad (1.1.11.1)$$

- Process Gain (from the FOPDT model in (1.1.11.1)):

$$K = \frac{\Delta y}{\Delta u} \quad (1.1.11.2)$$

- Maximum Slope:

$$R = \max \left( \frac{dy}{dt} \right) \quad (1.1.11.3)$$

- Dead Time: Using the maximum-slope construction in (1.1.11.3), if

$(t_{\text{inflection}}, y_{\text{inflection}})$  is the inflection point,

$$t_i = t_{\text{inflection}} - \frac{y_{\text{inflection}} - y_0}{R} \quad (1.1.11.4)$$

$$\theta = t_i - t_0 \quad (1.1.11.5)$$

- Time Constant: Using the process gain and maximum slope definitions in (1.1.11.2) and (1.1.11.3),

$$\tau = \frac{K\Delta u}{R} \quad (1.1.11.6)$$

### 1.1.12 Transfer Function and Feedback Loop

- Transfer Function:

$$G(s) = \frac{Y(s)}{X(s)} \quad (1.1.12.1)$$

$$Y(s) = G(s)X(s) \quad (1.1.12.2)$$

- Negative Feedback: For forward transfer function  $G(s)$  and feedback transfer function  $H(s)$ ,

$$E(s) = X(s) - H(s)Y(s) \quad (1.1.12.3)$$

$$Y(s) = G(s)E(s) \quad (1.1.12.4)$$

- Closed-Loop Transfer Function: Using the relations in (1.1.12.3) and (1.1.12.4),

$$Y(s) = G(s)[X(s) - H(s)Y(s)] \quad (1.1.12.5)$$

$$Y(s)[1 + G(s)H(s)] = G(s)X(s) \quad (1.1.12.6)$$

$$\frac{Y(s)}{X(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad (1.1.12.7)$$

- Unity Feedback: Setting the feedback transfer function to unity in (1.1.12.7),

$$H(s) = 1 \quad (1.1.12.8)$$

$$\frac{Y(s)}{X(s)} = \frac{G(s)}{1 + G(s)} \quad (1.1.12.9)$$

### 1.1.13 Routh-Hurwitz Stability Criterion

- Characteristic Equation: For a characteristic polynomial

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0 \quad (1.1.13.1)$$

- Routh Array:

$$\begin{pmatrix} a_n & a_{n-2} & a_{n-4} & \cdots \\ a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\ b_1 & b_2 & b_3 & \cdots \\ c_1 & c_2 & c_3 & \cdots \end{pmatrix} \quad (1.1.13.2)$$

- First Elements: From the Routh array structure in (1.1.13.2), the first elements are

$$b_1 = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} \quad (1.1.13.3)$$

$$b_2 = \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}} \quad (1.1.13.4)$$

- Number of Unstable Poles:

$$N_{\text{RHP}} = \text{number of sign changes in the first column} \quad (1.1.13.5)$$

- Stability Conditions:

$$\text{Stable system} \iff \text{no sign changes in the first column} \quad (1.1.13.6)$$

$$\text{Unstable system} \iff \text{one or more sign changes in the first column} \quad (1.1.13.7)$$

$$\text{Number of unstable poles} = N_{\text{RHP}} \quad (1.1.13.8)$$



First calculate the product:

$$z_1 z_2 = (1 - i)i \quad (1.2.2.4)$$

$$= i - i^2 \quad (1.2.2.5)$$

$$= 1 + i. \quad (1.2.2.6)$$

The complex number  $1 + i$  has equal positive real and imaginary parts. Hence,

$$\arg(1 + i) = \tan^{-1} \left( \frac{1}{1} \right) \quad (1.2.2.7)$$

$$= \frac{\pi}{4}. \quad (1.2.2.8)$$

Therefore,

$$\boxed{\arg(z_1 z_2) = \frac{\pi}{4}.} \quad (1.2.2.9)$$

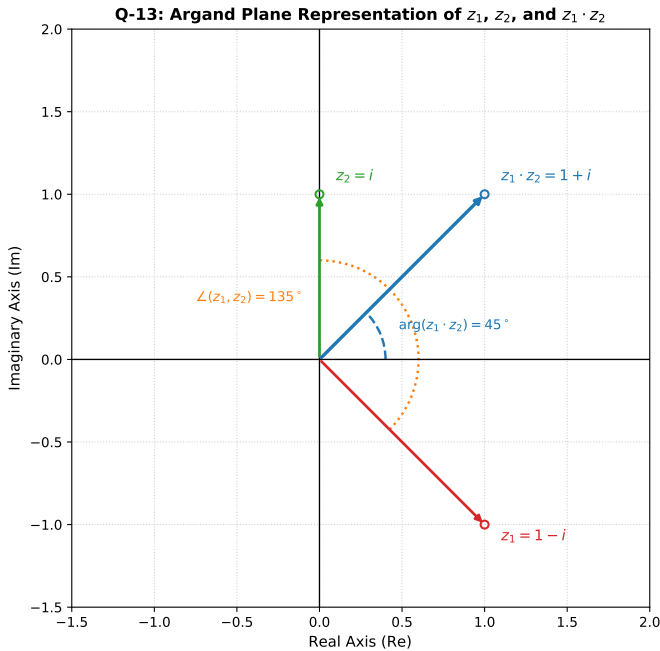


Fig. 1.2.2.1. Complex-plane representation of  $z_1 z_2$

codes/q13/main.py

1.2.3 A box contains 3 identical green balls and 7 identical blue balls. Two balls are randomly drawn without replacement from the box. The probability of drawing 1 green and 1 blue ball is (GATE CH 2025)

a)  $\frac{{}^3P_1 \times {}^7P_1}{{}^{10}P_2}$   
 b)  $\frac{{}^{10}P_3 \times {}^{10}P_7}{{}^{10}P_2}$

c)  $\frac{{}^{10}C_3 \times {}^{10}C_7}{{}^{10}C_2}$   
 d)  $\frac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2}$

**Solution:** There are

$$3 + 7 = 10 \quad (1.2.3.1)$$

balls in total. We need one green ball and one blue ball.

Using the combination formula in (1.1.4.1), the number of favourable selections is

$${}^3C_1 {}^7C_1. \quad (1.2.3.2)$$

The total number of ways of selecting two balls from ten balls is

$${}^{10}C_2. \quad (1.2.3.3)$$

Therefore,

$$P(1 \text{ green}, 1 \text{ blue}) = \frac{{}^3C_1 {}^7C_1}{{}^{10}C_2} \quad (1.2.3.4)$$

$$= \frac{3 \times 7}{\frac{10 \times 9}{2}} \quad (1.2.3.5)$$

$$= \frac{21}{45} \quad (1.2.3.6)$$

$$= \frac{7}{15}. \quad (1.2.3.7)$$

Thus,

$$P = \frac{7}{15}. \quad (1.2.3.8)$$

codes/q14/main.c

codes/q14/main.dat

1.2.4 The residence time distribution function  $E(t)$  for a reactor is given by

$$E(t) = \begin{cases} 1 - 2t, & t \leq 0.5 \text{ min}, \\ 0, & t > 0.5 \text{ min}. \end{cases} \quad (1.2.4.1)$$

The mean residence time (in min) is \_\_\_\_\_ (rounded off to two decimal places). (GATE CH 2025)

**Solution:** Using the mean residence-time relation in (1.1.3.1),

$$\bar{t} = \frac{\int_0^{\infty} tE(t) dt}{\int_0^{\infty} E(t) dt}. \quad (1.2.4.2)$$

Since  $E(t) = 0$  for  $t > 0.5$ ,

$$\bar{t} = \frac{\int_0^{0.5} t(1-2t) dt}{\int_0^{0.5} (1-2t) dt}. \quad (1.2.4.3)$$

For the numerator,

$$\int_0^{0.5} t(1-2t) dt = \int_0^{0.5} (t-2t^2) dt \quad (1.2.4.4)$$

$$= \left[ \frac{t^2}{2} - \frac{2t^3}{3} \right]_0^{0.5} \quad (1.2.4.5)$$

$$= \frac{1}{8} - \frac{1}{12} \quad (1.2.4.6)$$

$$= \frac{1}{24}. \quad (1.2.4.7)$$

For the denominator,

$$\int_0^{0.5} (1-2t) dt = [t-t^2]_0^{0.5} \quad (1.2.4.8)$$

$$= \frac{1}{2} - \frac{1}{4} \quad (1.2.4.9)$$

$$= \frac{1}{4}. \quad (1.2.4.10)$$

Hence,

$$\bar{t} = \frac{1/24}{1/4} = \frac{1}{6} = 0.1667 \text{ min} \quad (1.2.4.11)$$

$$\approx 0.17 \text{ min.} \quad (1.2.4.12)$$



Here,

$$v_x = ax, \quad v_y = -by. \quad (1.2.5.6)$$

Therefore,

$$\frac{\partial v_x}{\partial x} = a, \quad \frac{\partial v_y}{\partial y} = -b. \quad (1.2.5.7)$$

Thus,

$$\nabla \cdot \mathbf{v} = a - b. \quad (1.2.5.8)$$

For zero divergence,

$$a - b = 0. \quad (1.2.5.9)$$

Therefore,  $a - b = 0$ .

codes/q36/main.py

1.2.6 Let  $X$  be a continuous random variable with probability density function

$$p(x) = \begin{cases} \frac{1}{a}, & x \in (0, a), \\ 0, & \text{otherwise,} \end{cases} \quad (1.2.6.1)$$

where  $a$  is a positive constant. Let  $f(x) = x^2$ . The expected value of  $f(X)$  is (GATE CH 2025)

- |                    |                     |
|--------------------|---------------------|
| a) $\frac{a^2}{3}$ | c) $\frac{2a^2}{3}$ |
| b) $\frac{a^3}{3}$ | d) $\frac{2a^3}{3}$ |

**Solution:** The given function and probability density are

$$f(x) = x^2, \quad (1.2.6.2)$$

$$p(x) = \frac{1}{a}, \quad 0 < x < a. \quad (1.2.6.3)$$

The expected value of a function of a continuous random variable is obtained from (1.1.5.1), which is the continuous form of taking a weighted average over all possible outcomes:

$$E[f(X)] = \int_{-\infty}^{\infty} f(x)p(x) dx. \quad (1.2.6.4)$$

Since the density is non-zero only from 0 to  $a$ ,

$$E[x^2] = \int_0^a x^2 \frac{1}{a} dx \quad (1.2.6.5)$$

$$= \frac{1}{a} \left[ \frac{x^3}{3} \right]_0^a \quad (1.2.6.6)$$

$$= \frac{1}{a} \frac{a^3}{3} \quad (1.2.6.7)$$

$$= \frac{a^2}{3}. \quad (1.2.6.8)$$

Therefore,

$$E[f(X)] = \frac{a^2}{3}. \quad (1.2.6.9)$$

The expected value represents the average output obtained when the possible outcomes are weighted according to their probabilities.

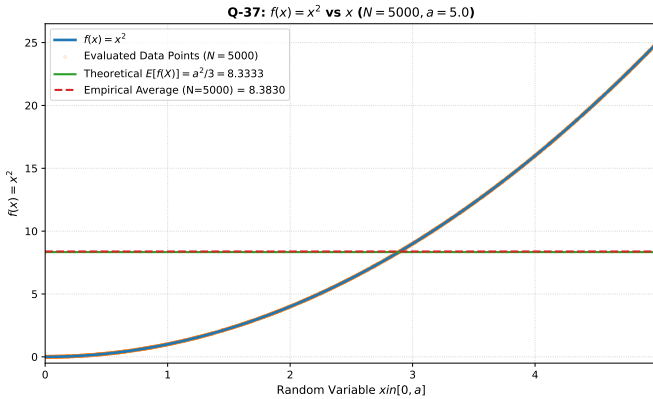


Fig. 1.2.6.1. Uniform probability density over the interval  $(0, a)$

figs/q37/Expected-Value.pdf

codes/q37/main.dat

codes/q37/main.py

codes/q37/main.c

### 1.2.7 The differential equation

$$\frac{dy}{dx} + \frac{y}{x} = 0 \quad (1.2.7.1)$$

has which of the following correct solution(s)?

(GATE CH 2025)

a)  $y = x + c$   
 b)  $y = \frac{c}{x}$

c)  $y = -x + c$   
 d)  $y = 0$

**Solution:** Rearranging the differential equation,

$$\frac{dy}{dx} = -\frac{y}{x}. \quad (1.2.7.2)$$

For numerical plotting, the trapezoidal method approximates the solution using the average slope over a step. For  $f(x, y) = -y/x$ , the update is

$$y_{n+1} = y_n + \frac{h}{2} \left[ -\frac{y_n}{x_n} - \frac{y_{n+1}}{x_{n+1}} \right]. \quad (1.2.7.3)$$

The analytical solution is obtained by separation of variables:

$$\frac{1}{y} dy = -\frac{1}{x} dx. \quad (1.2.7.4)$$

Integrating,

$$\ln |y| = -\ln |x| + C, \quad (1.2.7.5)$$

$$\ln |xy| = C, \quad (1.2.7.6)$$

$$y = \frac{C}{x}. \quad (1.2.7.7)$$

For  $C = 0$ ,

$$y = 0. \quad (1.2.7.8)$$

Hence, the correct solutions are

$$\boxed{\text{(B) and (D)}}. \quad (1.2.7.9)$$

The trapezoidal method can then be used to generate numerical points for plotting the solution family.

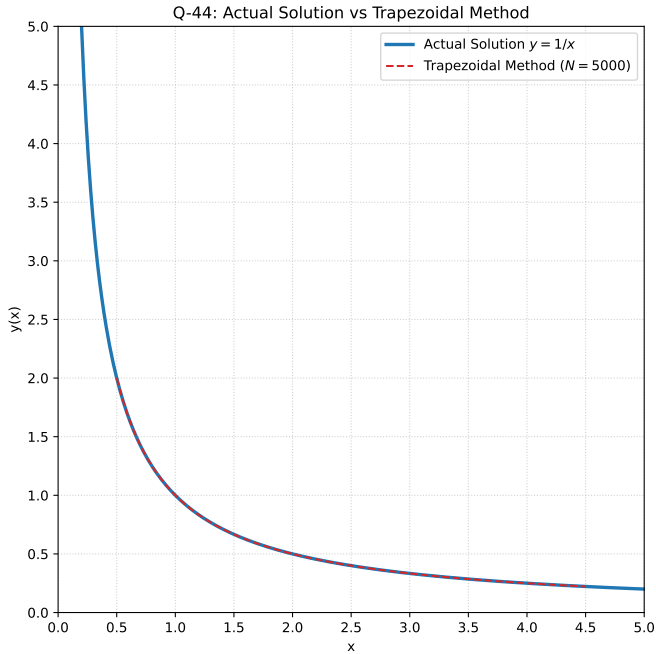


Fig. 1.2.7.1. Solution family  $y = C/x$  for the given differential equation

figs/q44/Diff-Eqn.pdf

codes/q44/main.py

1.2.8 Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}. \quad (1.2.8.1)$$

One of its eigenvalues is 0.27. The other eigenvalue, rounded off to two decimal places, is \_\_\_\_\_. (GATE CH 2025)

**Solution:** For an eigenvalue  $\lambda$  and its corresponding eigenvector  $\mathbf{x}$ ,

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}, \quad \mathbf{x} \neq \mathbf{0}. \quad (1.2.8.2)$$

The eigenvalues are obtained from the characteristic equation

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0. \quad (1.2.8.3)$$

For the given matrix,

$$\det \begin{pmatrix} 2 - \lambda & 3 \\ 1 & 2 - \lambda \end{pmatrix} = 0. \quad (1.2.8.4)$$

Therefore,

$$(2 - \lambda)^2 - 3 = 0 \quad (1.2.8.5)$$

$$\lambda^2 - 4\lambda + 1 = 0. \quad (1.2.8.6)$$

Solving the quadratic equation,

$$\lambda = \frac{4 \pm \sqrt{16 - 4}}{2} \quad (1.2.8.7)$$

$$= 2 \pm \sqrt{3}. \quad (1.2.8.8)$$

Thus,

$$\lambda_1 = 2 - \sqrt{3} \approx 0.27, \quad (1.2.8.9)$$

$$\lambda_2 = 2 + \sqrt{3} \approx 3.73. \quad (1.2.8.10)$$

For completeness, the corresponding eigenvectors can be obtained from  $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0$ . For  $\lambda_1 = 2 - \sqrt{3}$ ,

$$\sqrt{3}x + 3y = 0, \quad (1.2.8.11)$$

so one eigenvector is

$$\mathbf{x}_1 = \begin{pmatrix} \sqrt{3} \\ -1 \end{pmatrix}. \quad (1.2.8.12)$$

For  $\lambda_2 = 2 + \sqrt{3}$ ,

$$-\sqrt{3}x + 3y = 0, \quad (1.2.8.13)$$

so one eigenvector is

$$\mathbf{x}_2 = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}. \quad (1.2.8.14)$$

From the trace relation in (1.1.7.3), the sum of the eigenvalues is

$$\lambda_1 + \lambda_2 = (2 - \sqrt{3}) + (2 + \sqrt{3}) \quad (1.2.8.15)$$

$$= 4. \quad (1.2.8.16)$$

The trace of the matrix is

$$\text{tr}(\mathbf{A}) = 2 + 2 = 4. \quad (1.2.8.17)$$

Hence,

$$\boxed{\lambda_1 + \lambda_2 = \text{tr}(\mathbf{A})}. \quad (1.2.8.18)$$

Using the given value  $\lambda_1 = 0.27$  and the trace,

$$\lambda_2 = 4 - 0.27 \quad (1.2.8.19)$$

$$= 3.73. \quad (1.2.8.20)$$

Therefore, the required value is

$$\boxed{3.73}. \quad (1.2.8.21)$$

codes/q45/main.py

1.2.9 The Newton-Raphson method is used to obtain the root of

$$f(x) = x^2 - x - 1 = 0 \quad (1.2.9.1)$$

with the initial guess  $x^{(0)} = 1$ . The value of the second iterate  $x^{(2)}$ , rounded off to two decimal places, is \_\_\_\_\_. (GATE CH 2025)

**Solution:** Newton-Raphson improves an initial guess by using the tangent to the function at the current point. Repeating the iteration moves the approximation towards the root.

The Newton-Raphson formula is given by (1.1.6.1):

$$x^{(n+1)} = x^{(n)} - \frac{f(x^{(n)})}{f'(x^{(n)})}. \quad (1.2.9.2)$$

Here,

$$f(x) = x^2 - x - 1, \quad (1.2.9.3)$$

$$f'(x) = 2x - 1. \quad (1.2.9.4)$$

Given

$$x^{(0)} = 1. \quad (1.2.9.5)$$

For the first iteration,

$$f(1) = -1, \quad (1.2.9.6)$$

$$f'(1) = 1, \quad (1.2.9.7)$$

and hence

$$x^{(1)} = 1 - \frac{-1}{1} = 2. \quad (1.2.9.8)$$

For the second iteration,

$$f(2) = 4 - 2 - 1 = 1, \quad (1.2.9.9)$$

$$f'(2) = 4 - 1 = 3. \quad (1.2.9.10)$$

Therefore,

$$x^{(2)} = 2 - \frac{1}{3} \quad (1.2.9.11)$$

$$= \frac{5}{3} \quad (1.2.9.12)$$

$$= 1.6667 \dots \quad (1.2.9.13)$$

Thus, rounded off to two decimal places,

$$\boxed{x^{(2)} = 1.67}. \quad (1.2.9.14)$$

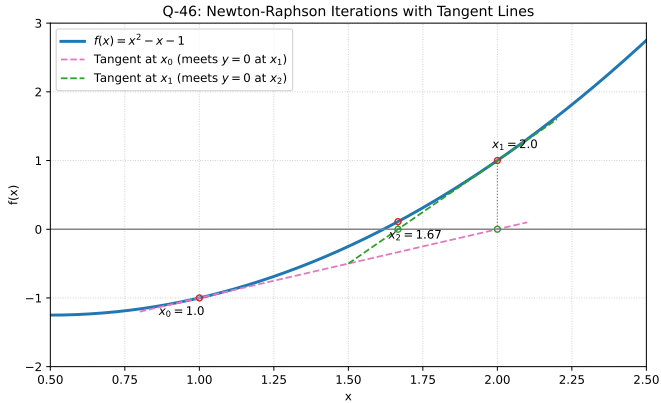


Fig. 1.2.9.1. Newton-Raphson iteration for  $f(x) = x^2 - x - 1$

figs/q46/Newton-Raphson.pdf

codes/q46/main.py

1.2.10 The transfer function of a process is

$$G_p(s) = \frac{2e^{-s}}{(5s + 1)^2}. \quad (1.2.10.1)$$

The process is fitted to a first-order plus dead-time (FOPDT) model using the maximum slope method. If  $\tau_m$  and  $\theta_m$  are the fitted time constant and dead time, respectively, then  $\tau_m/\theta_m$ , rounded off to two decimal places, is \_\_\_\_\_. (GATE CH 2025)

The unit-step response and its derivatives for  $1/(\tau s + 1)^2$  are

$$y(t) = 1 - \left(1 + \frac{t}{\tau}\right) e^{-t/\tau}, \quad (1.2.10.2)$$

$$\frac{dy}{dt} = \frac{t}{\tau^2} e^{-t/\tau}, \quad (1.2.10.3)$$

$$\frac{d^2y}{dt^2} = \frac{1}{\tau^2} \left(1 - \frac{t}{\tau}\right) e^{-t/\tau}. \quad (1.2.10.4)$$

**Solution:** From the time-delay relation, the factor  $e^{-s}$  represents a pure time delay of 1 min. Thus the dynamic response of the second-order part is shifted by 1 min.

For the dynamic part,

$$G_0(s) = \frac{2}{(5s + 1)^2}, \quad (1.2.10.5)$$

so

$$K = 2, \quad \tau = 5 \text{ min}, \quad \theta = 1 \text{ min}. \quad (1.2.10.6)$$

For the normalized second-order response, the maximum slope occurs at the inflection point, where

$$\frac{d^2y}{dt^2} = 0. \quad (1.2.10.7)$$

Hence,

$$1 - \frac{t}{\tau} = 0 \implies t = \tau = 5 \text{ min}. \quad (1.2.10.8)$$

Including the dead time,

$$t_{\text{inflection}} = 1 + 5 = 6 \text{ min}. \quad (1.2.10.9)$$

At this point, the output is

$$y_{\text{inflection}} = 2 \left[ 1 - \left( 1 + \frac{5}{5} \right) e^{-1} \right] \quad (1.2.10.10)$$

$$= 2 \left( 1 - \frac{2}{e} \right). \quad (1.2.10.11)$$

The maximum slope is

$$R = 2 \left[ \frac{t}{\tau^2} e^{-t/\tau} \right]_{t=5} \quad (1.2.10.12)$$

$$= \frac{2}{5e}. \quad (1.2.10.13)$$

Using the maximum-slope relations in (1.1.11.3) and (1.1.11.6), the fitted time constant is

$$\tau_m = \frac{K \Delta u}{R} \quad (1.2.10.14)$$

$$= \frac{2 \times 1}{2/(5e)} \quad (1.2.10.15)$$

$$= 5e. \quad (1.2.10.16)$$

Using the dead-time tangent relation in (1.1.11.4), the tangent at the inflection point

intersects the time axis at

$$\theta_m = t_{\text{inflection}} - \frac{y_{\text{inflection}}}{R} \quad (1.2.10.17)$$

$$= 6 - \frac{2(1 - 2/e)}{2/(5e)} \quad (1.2.10.18)$$

$$= 16 - 5e. \quad (1.2.10.19)$$

Therefore,

$$\frac{\tau_m}{\theta_m} = \frac{5e}{16 - 5e} \quad (1.2.10.20)$$

$$\approx 5.64. \quad (1.2.10.21)$$

Thus,

$$\boxed{\frac{\tau_m}{\theta_m} = 5.64}. \quad (1.2.10.22)$$

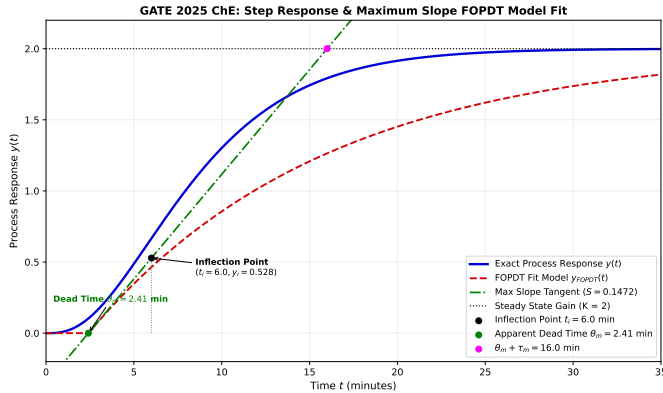


Fig. 1.2.10.1. Maximum-slope FOPDT approximation of the given process response

figs/q60/FOPDT-Modelling.pdf

codes/q60/main.py

1.2.11 The block diagram of a series cascade control system (with time in minutes) is shown in the figure. For  $\tau_I = 8$  min and  $K_c^s = 1$ , the maximum value of  $K_c^m$ , below which the cascade control system is stable, is \_\_\_\_\_ (rounded off to the nearest integer). (GATE CH 2025)

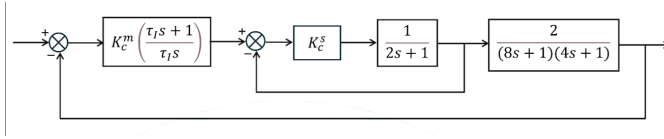


Fig. 1.2.11.1. Series cascade control system given in GATE CH 2025 Q62

**Solution:** The inner process is

$$G_1(s) = \frac{1}{2s + 1}, \quad (1.2.11.1)$$

and the inner controller gain is  $K_c^s = 1$ . Therefore, the closed-loop inner transfer function is

$$G_{\text{inner}}(s) = \frac{K_c^s G_1(s)}{1 + K_c^s G_1(s)} \quad (1.2.11.2)$$

$$= \frac{1/(2s + 1)}{1 + 1/(2s + 1)} \quad (1.2.11.3)$$

$$= \frac{1}{2(s + 1)}. \quad (1.2.11.4)$$

The outer process is

$$G_2(s) = \frac{2}{(8s + 1)(4s + 1)}. \quad (1.2.11.5)$$

Hence the equivalent process for the outer loop is

$$G_{\text{eq}}(s) = G_{\text{inner}}(s) G_2(s) \quad (1.2.11.6)$$

$$= \frac{1}{(s + 1)(8s + 1)(4s + 1)}. \quad (1.2.11.7)$$

The outer controller is

$$G_c(s) = K_c^m \frac{\tau_I s + 1}{\tau_I s}. \quad (1.2.11.8)$$

Since  $\tau_I = 8$  min,

$$G_c(s) = K_c^m \frac{8s + 1}{8s}. \quad (1.2.11.9)$$

Therefore, the open-loop transfer function of the outer loop is

$$G_c(s) G_{\text{eq}}(s) = K_c^m \frac{8s + 1}{8s(s + 1)(8s + 1)(4s + 1)} \quad (1.2.11.10)$$

$$= \frac{K_c^m}{8s(s + 1)(4s + 1)}. \quad (1.2.11.11)$$

For unity negative feedback, using (1.1.12.9), the characteristic equation is

$$1 + \frac{K_c^m}{8s(s + 1)(4s + 1)} = 0, \quad (1.2.11.12)$$

which gives

$$8s(s+1)(4s+1) + K_c^m = 0 \quad (1.2.11.13)$$

$$32s^3 + 40s^2 + 8s + K_c^m = 0. \quad (1.2.11.14)$$

Using the Routh-Hurwitz relations in (1.1.13.2), the Routh array is

$$\begin{pmatrix} 32 & 8 \\ 40 & K_c^m \\ \frac{40(8) - 32K_c^m}{40} & 0 \\ \frac{40}{K_c^m} & \end{pmatrix}. \quad (1.2.11.15)$$

For stability, every element in the first column must be positive. Thus,

$$8 - 0.8K_c^m > 0, \quad (1.2.11.16)$$

$$K_c^m > 0. \quad (1.2.11.17)$$

Therefore,

$$0 < K_c^m < 10. \quad (1.2.11.18)$$

Hence the maximum value of  $k_c^m$  below which the system is stable, rounded to the nearest integer, is 10.

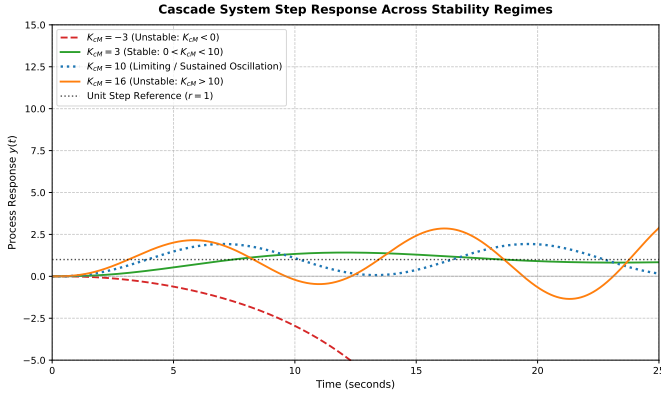


Fig. 1.2.11.2. Cascade-loop reduction and Routh-Hurwitz stability calculation for Q62

<figs/q62/Response-Plot.pdf>

<codes/q62/main.py>